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PUMPING TESTS IN NONUNIFORM AQUIFERS — THE RADially SYMMETRIC CASE

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ABSTRACT

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Traditionally, pumping-test-analysis methodology has been limited to applications involving aquifers whose properties are assumed uniform in space. This work attempts to assess the applicability of analytical methodology to a broader class of units with spatially varying properties. An examination of flow behavior in a simple configuration consisting of pumping from the center of a circular disk embedded in a matrix of differing properties is the basis for this investigation. A solution describing flow in this configuration is obtained through Laplace-transform techniques using analytical and numerical inversion schemes. Approaches for the calculation of flow properties in conditions that can be roughly represented by this simple configuration are proposed. Possible applications include a wide variety of geologic structures, as well as the case of a well skin resulting from drilling or development. Of more importance than the specifics of these techniques for analysis of water-level responses is the insight into flow behavior during a pumping test that is provided by the large-time form of the derived solution. The solution reveals that drawdown during a pumping test can be considered to consist of two components that are dependent and independent of near-well properties, respectively. Such an interpretation of pumping-test drawdown allows some general conclusions to be drawn concerning the relationship between parameters calculated using analytical approaches based on curve-matching and those calculated using approaches based on the slope of a semilog straight line plot. The infinite-series truncation that underlies the semilog analytical approaches is shown to remove further contributions of near-well material to total drawdown. In addition, the semilog distance–drawdown approach is shown to yield an expression that is equivalent to the Thiem equation. These results allow some general recommendations to be made concerning observation-well placement for pumping tests in nonuniform aquifers. The relative diffusivity of material on either side of a discontinuity is shown to be the major factor in controlling flow behavior during the period in which the front of the cone of depression is moving across the discontinuity. Though resulting from an analysis of flow in an idealized configuration, the insights of this work into flow behavior during a pumping test are applicable to a wide class of nonuniform units.

INTRODUCTION

The pumping test has traditionally been the standard method used to evaluate the transmissive and storage properties of subsurface material for

water-supply purposes. Response data from a pumping test can be employed to estimate subsurface properties using inversing schemes based on analytical or numerical solutions to the appropriate flow conditions. Due to developments before the widespread use of computers and the normal sparsity of observation-well data, inversing schemes based on analytical solutions have primarily been used for evaluation of system properties. This methodology is founded on a series of assumptions required to reduce the complex natural system to a mathematically tractable one. The field of well hydraulics has developed around the derivation of analytical solutions for conditions in which various of these assumptions concerning the flow system have been relaxed (Wenzel, 1942; Hantush, 1964). Recent work (Butler, 1986) has examined the ramifications of one major assumption; namely, that the transmissive properties of a confined aquifer are invariant in space. That work, however, was constrained to an analysis of behavior in units whose property variations were of a relatively smoothly varying nature in the horizontal plane, the nonuniform structure being characterized by correlated elements drawn from a second-order stationary stochastic process, i.e. a single-component system. Further work is required to examine the question of flow behavior under conditions of more abrupt variations in the horizontal plane, i.e. multicomponent systems.

Since all aquifers in the field are inherently nonuniform, the question of the ramifications of the uniformity assumption is one of considerable concern. Specifically, the issue of whether much worthwhile information about the flow system can be obtained from a pumping-test analysis using a few observation wells must be addressed. Clearly, the physical meaning of the parameters determined from a pumping-test analysis in a nonuniform aquifer requires examination. Is a pumping-test parameter some type of average of all material sensed during pumping, or can more detailed information about the nonuniform nature of a unit be obtained from the analysis? Butler (1986) has partly addressed these issues, but further work is required to support and extend his conclusions.

To date, relatively little work has touched upon the theory of pumping-test-analysis methodology in nonuniform aquifers. In terms of numerical studies, Butler (1986) presented an analysis employing both deterministic and stochastic models of permeability variations. Earlier work includes an examination of pressure build-up tests in units of nonuniform permeability by Warren and Price (1961) and an examination of pumping tests in aquifers of uniform diffusivity but variable thickness by Vandenberg (1977). A number of workers have also employed numerical approaches to develop type curves for pumping tests in simple nonuniform aquifers (e.g., Bixel and Van Poollen, 1967; Kazemi et al., 1972; Merrill et al., 1974).

In terms of analytical approaches, most work has concentrated on units composed of two components separated by a linear or radial discontinuity. An

exception to this is the analysis of Hantush (1962) of behavior in a perfectly confined aquifer of exponentially increasing thickness. Bixel et al. (1963) developed a solution for the case of a single linear discontinuity in a horizontal plane using Laplace and exponential Fourier transforms. The complexity of their solution, however, limited evaluation to a select number of cases. Nind (1965) and Fenske (1984) addressed the same question using image-well theory. A large body of work considers pumping tests in multilayered systems in which properties are uniform within a layer, but vary across layer boundaries. Streltsova (1978, 1984) provides summaries of work involving a single linear discontinuity in the vertical plane. Recent extensions to multilayered cases include Wikramaratna (1984) and Hunt (1985).

A number of workers have examined the effect of radial discontinuities on flow behavior during a pumping test using analytical approaches. Most of this work is found in the petroleum-engineering literature due to the common occurrence of such a configuration in petroleum applications. Hurst (1960) obtained a solution to the case of pumping from a radial disk embedded in an infinite unit of differing properties using Laplace-transform techniques. The complexity of the Laplace-space expression, however, resulted in only a large-time solution being presented. Since that initial study, a number of other workers have attempted to find analytical solutions to this flow configuration (e.g., Loucks and Guerrero, 1961; Carter, 1966; Odeh, 1969). Apparently, few workers have addressed this question in the groundwater literature. Barker and Herbert (1982) attempted a Laplace-transform-based analytic solution for flow in what they termed "patchy" aquifers. Once again, the complexity of the solution in Laplace space resulted in only a large-time solution being proposed. Şen (1987) employed an analysis based on the Boltzmann transformation to propose type curves for all times of pumpage. It should be noted that all of these authors were examining the question of pumping from the center of a disk embedded in a uniform matrix; i.e. the radially symmetric case. Larkin (1963) presented a large-time solution for pumping anywhere in the circular disk.

The issue of pumping tests in the presence of radially symmetric discontinuities will be reexamined in this article. Flow behavior in this configuration is of importance for several reasons. First, the configuration can be employed to conceptualize a wide variety of geologic settings where a "patch" of material is embedded in a matrix of differing permeability. An example of such a geologic setting is given by Barker and Herbert (1982). Another common example is that of a cutoff meander channel in a flood-plain matrix. The oft-noted dependence of aquifer parameters on drilling methodology (e.g., McKown et al., 1987) is yet a further example of a condition that can be addressed using this configuration. Second, the radially symmetric configuration presents a structure that is accessible to analytical analysis, thus allowing the significance of property variations to be readily evaluated. Third, and of greatest importance, the insight that can be gained from the nature of this

analytic solution enables the development of a better general understanding of pumping tests in nonuniform units.

Unlike the emphasis of earlier work, this article will focus on the implications of the derived solution for pumping tests in general. By employing a configuration in which flow properties vary simply with radial distance from the pumping well, much can be learned concerning the portions of the aquifer that control drawdown at different times during a pumping test. Consideration of the controlling influences on drawdown at any given time allows a pumping test to be a more valuable tool in assessing the nature of property variations within a nonuniform unit. An analysis of flow behavior in a two-component system, in which properties within each component are uniform, can be justified in that Butler (1986) has shown that property variations within single-component systems can often be ignored. Therefore, the variations between components should be the focus of interest.

The approach of this article will be to begin by deriving a simple analytic solution to the radial-flow problem by using a straightforward Laplace-transform procedure. The solution that will be derived in Laplace space can only be readily analytically inverted to untransformed space at relatively large times, so the Stehfest algorithm (Stehfest, 1970) will be employed to evaluate the less readily inverted portions of the solutions. Implementation of both the large-time and numerically inverted solutions for analyses of pumping tests in the presence of a radially symmetric discontinuity will be considered. The form of the large-time solution will then serve as the basis for a discussion of the implications for pumping tests in general.

PROBLEM STATEMENT

The problem of interest here is that of pumping from a fully penetrating well at the center of a circular disk of radius R , which is embedded in an infinite

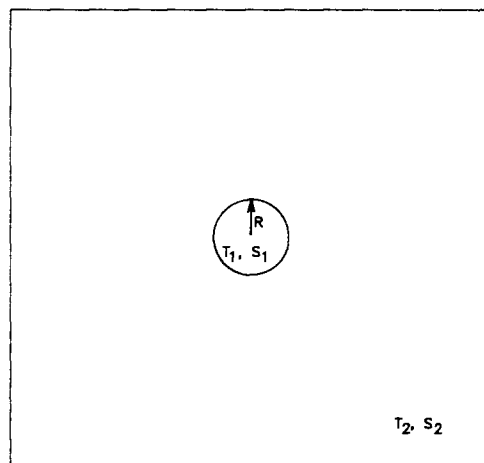


Fig. 1. Configuration consisting of a disk of material 1 (radius = R) embedded in matrix of material 2.

matrix (Fig. 1). Flow properties are assumed uniform within both the disk and the matrix, although differing between the two. Save for the variation of properties between the disk and matrix, the standard conditions for a pumping test in a confined aquifer are assumed at all times. Flow within the circular disk can be described mathematically by the radial-flow equation:

$$\frac{\partial^2 s_1}{\partial r^2} + \frac{1}{r} \frac{\partial s_1}{\partial r} = \frac{S_1}{T_1} \frac{\partial s_1}{\partial t} \quad (1)$$

where: s_i = drawdown in material i , [L]; r = radial direction [L]; S_i = storage coefficient of material i , dimensionless; and T_i = transmissivity of material i [L^2/T]. Flow in the matrix can be described by an expression analogous to (1):

$$\frac{\partial^2 s_2}{\partial r^2} + \frac{1}{r} \frac{\partial s_2}{\partial r} = \frac{S_2}{T_2} \frac{\partial s_2}{\partial t} \quad (2)$$

Boundary and initial conditions are as follows:

$$\lim_{r_w \rightarrow 0} 2\pi r_w T_1 \frac{\partial s_1}{\partial r} = -Q \quad (3)$$

$$s_2(\infty, t) = 0 \quad (4)$$

$$s_1(r, 0) = s_2(r, 0) = 0, \quad r_w < r < \infty \quad (5)$$

where: Q = pumpage from well, [L^3/T]; and r_w = radius of pumping well, [L]. In order to ensure continuity of flow between the disk and the matrix, auxiliary conditions at the matrix-disk boundary (R) must also be met:

$$s_1(R, t) = s_2(R, t) \quad (6)$$

$$T_1 \frac{\partial s_1(R, t)}{\partial r} = T_2 \frac{\partial s_2(R, t)}{\partial r} \quad (7)$$

Derivation of analytical solution

Equations (1)–(7) describe the flow conditions of interest here. In order to find the functions s_1 and s_2 that satisfy these conditions for any given Q , R , T_i , and S_i , the Laplace-transformation technique can be applied. The Laplace-transformation approach (Carslaw and Jaeger, 1959; Hantush, 1959, 1964) enables a system of partial differential equations in r and t to be transformed into a system of ordinary differential equations in r . The transformed equivalents of (1)–(7) are given below:

$$\frac{\partial^2 \bar{s}_1}{\partial r^2} + \frac{1}{r} \frac{\partial \bar{s}_1}{\partial r} = \frac{S_1}{T_1} p \bar{s}_1 \quad (8)$$

$$\frac{\partial^2 \bar{s}_2}{\partial r^2} + \frac{1}{r} \frac{\partial \bar{s}_2}{\partial r} = \frac{S_2}{T_2} p \bar{s}_2 \quad (9)$$

$$\lim_{r_w \rightarrow 0} 2\pi r_w T_1 \frac{\partial \bar{s}_1}{\partial r} = -Q/p \quad (10)$$

$$\bar{s}_2(\infty, p) = 0 \quad (11)$$

$$\bar{s}_1(R, p) = \bar{s}_2(R, p) \quad (12)$$

$$T_1 \frac{\partial \bar{s}_1(R, p)}{\partial r} = T_2 \frac{\partial \bar{s}_2(R, p)}{\partial r} \quad (13)$$

where: $\bar{s}_i$ = the Laplace transform of s_i ; and p = Laplace-transform variable.

The solution to this system of transformed equations is quite straightforward as eqns. (8) and (9) are simply forms of the modified Bessel equation (Bronshtein and Semendyayev, 1985). A solution can therefore be proposed in the form:

$$\bar{s}_1 = C_1 K_0(Nr) + C_2 I_0(Nr) \quad (14)$$

$$\bar{s}_2 = C_3 K_0(Ar) + C_4 I_0(Ar) \quad (15)$$

where: I_i = modified Bessel function of the first kind of order i ; K_i = modified Bessel function of the second kind of order i ; $N = \sqrt{S_1 p / T_1}$; $A = \sqrt{S_2 p / T_2}$; and C_i = constant.

Using the boundary conditions (10)–(13), the constants in eqns. (14) and (15) can be found resulting in the following expressions:

$$\bar{s}_1 = \left(\frac{Q}{2\pi T_1} \right) \frac{K_0(Nr)}{p} + \left(\frac{Q}{2\pi T_1} \right) \times \frac{\left[K_1(NR) K_0(AR) - \frac{T_2}{T_1} \frac{A}{N} K_0(NR) K_1(AR) \right] I_0(Nr)}{p \left[\frac{T_2}{T_1} \frac{A}{N} I_0(NR) K_1(AR) + I_1(NR) K_0(AR) \right]} \quad (16)$$

$$\bar{s}_2 = \left(\frac{Q}{2\pi T_1} \right) \frac{\left[K_0(NR) I_1(NR) + K_1(NR) I_0(NR) \right]}{p \left[\frac{T_2}{T_1} \frac{A}{N} I_0(NR) K_1(AR) + I_1(NR) K_0(AR) \right]} K_0(Ar) \quad (17)$$

Note that when $T_1 = T_2 = T$ and $S_1 = S_2 = S$, the two equations reduce to:

$$\bar{s}_1 = \left(\frac{Q}{2\pi T} \right) \frac{K_0\left(\sqrt{\frac{Sp}{T}} r\right)}{p} = \bar{s}_2 \quad (18)$$

which is simply the Laplace-space form of the solution of Theis (1935). Also note that eqns. (16) and (17) differ from eqns. (A.8) and (A.9) of Barker and Herbert (1982), which apparently have been reduced by ignoring several terms.

The final step of the Laplace-transformation approach is to invert back to

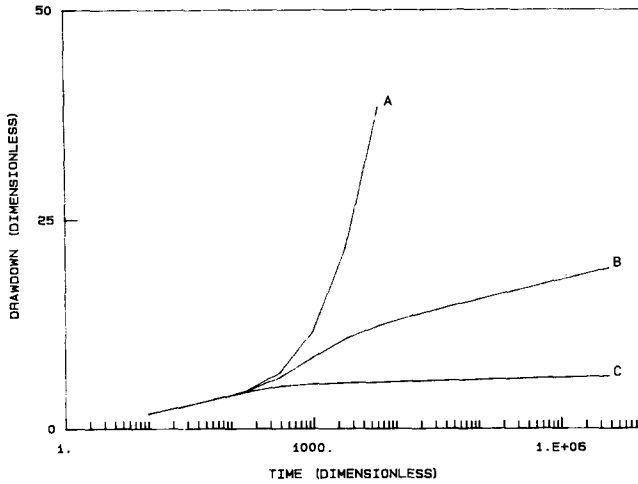


Fig. 2. Dimensionless drawdown ($4\pi s_1 T_1/Q$) versus time ($4T_1 t/S_1 r^2$) plot of numerically inverted solution ($R/r = 10$, $S_2/S_1 = 0.01$, $T_{2A}/T_1 = 0.1$, $T_{2B}/T_1 = 1$, $T_{2C}/T_1 = 10$).

untransformed space. For equations of the complexity of (16) and (17), an analytic transformation is only readily performed under quite limited conditions. In the general case, the inversion is best performed numerically. An algorithm developed by Stehfest (1970), which has been found to be of great use in hydrologic applications (Moench and Ogata, 1981, 1984), was employed here. Figure 2 displays the inverted solution of eqn. (16) for a set of example conditions. Note that the solution consists of two approximately straight line segments on a semilog plot separated by a transition zone. In this example, the

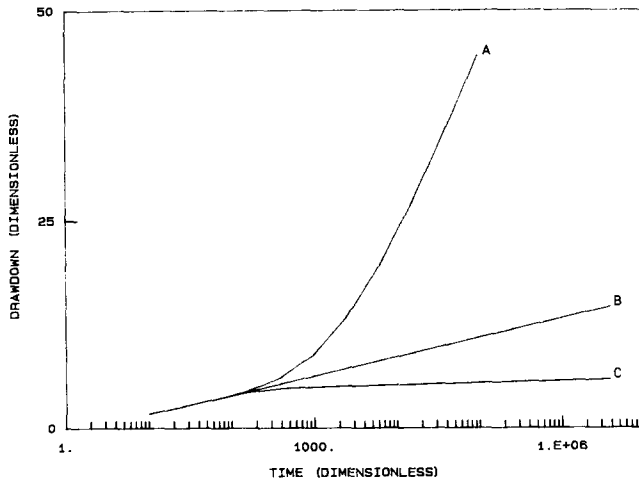


Fig. 3. Dimensionless drawdown versus time plot of numerically inverted solution ($R/r = 10$, $S_2/S_1 = 1$, $T_{2A}/T_1 = 0.1$, $T_{2B}/T_1 = 1$, $T_{2C}/T_1 = 10$).

difference between the storage coefficients of the two materials has been set to a large value in order to accentuate the transition. Figure 3 displays the numerically inverted solution for conditions equivalent to those of Fig. 2, save for the two storage coefficients being equal. The straight line that would be predicted from the Cooper-Jacob equation (Cooper and Jacob, 1946) for an infinite uniform aquifer appears as the middle line of Fig. 3.

The linear form displayed by portions of the numerically inverted solution on Figs. 2 and 3 suggests that a Cooper-Jacob analysis may be applicable for the radially symmetric nonuniform case. In order to explore this possibility further, the large-time form of the solution must be examined. It can be shown through analytic transformation of eqn. (18) that an argument of less than 0.1 for K_0 is approximately equivalent to the $u < 0.01$ criterion commonly employed to reduce the solution of Theis (1935) to the truncated-series form proposed by Cooper and Jacob (1946). When an argument z is less than 0.1, the modified Bessel functions can be reduced to the following approximate forms:

$$\begin{aligned} I_0(z) &= 1, I_1(z) = z/2, \\ K_0(z) &= \ln(2/Cz), K_1(z) = 1/z \end{aligned} \quad (19)$$

where $C = \exp \gamma$, $\gamma = \text{Euler's constant (0.5772...)}$.

Assuming that NR , Nr , AR , and $Ar < 0.1$, the relationships of (19) can be utilized to simplify eqns. (16) and (17) to the following forms:

$$\bar{s}_1 = \left(\frac{Q}{2\pi T_1 p} \right) \ln \left(\frac{2}{CNR} \right) + \left(\frac{Q}{2\pi T_1 p} \right) \left[\frac{\frac{1}{NR} \ln \left(\frac{2}{CAR} \right) + \frac{T_2}{T_1} \frac{\ln \left(\frac{CNR}{2} \right)}{NR}}{\frac{T_2}{T_1} \frac{1}{NR} + \frac{NR}{2} \ln \left(\frac{2}{CAR} \right)} \right] \quad (20)$$

$$\bar{s}_2 = \left(\frac{Q}{2\pi T_1 p} \right) \left[\frac{\frac{NR}{2} \ln \left(\frac{2}{CNR} \right) + \frac{1}{NR}}{\frac{T_2}{T_1} \frac{1}{NR} + \frac{NR}{2} \ln \left(\frac{2}{CAR} \right)} \right] \ln \left(\frac{2}{CAR} \right) \quad (21)$$

It can be shown under most conditions that the denominator of the bracketed expressions of eqns. (20) and (21) can be approximated by $T_2/(T_1 NR)$. In the case of eqn. (20), this will allow reduction to a form that can be analytically inverted to the expression given below:

$$s_1 = \frac{Q}{4\pi T_2} \ln \left(\frac{4T_2 t}{CS_2 R^2} \right) + \frac{Q}{2\pi T_1} \ln \frac{R}{r} \quad (22)$$

Equation (22) is equivalent to the large-time solution (A.12) given by Barker and Herbert (1982). The form of eqn. (22) is worthy of further consideration. Note that a plot of the logarithm of time versus drawdown will produce a straight line, as in the uniform-aquifer case. Application of the Cooper-Jacob approach for

pumping-test analysis, using the change in drawdown over a single $\log_{10}$ cycle in time (Δs), will then yield the following expression for transmissivity:

$$T_2 = 2.3Q/(4\pi\Delta s) \quad (23)$$

It is of interest to note that the calculated transmissivity is that of the matrix, and not the disk that contains both the pumping and observation wells. After a certain time, therefore, the properties of the disk have effectively no influence on the change in drawdown. However, as eqn. (22) shows, the total drawdown is always a function of the properties of both the disk and the matrix. This result is in keeping with the hypothesis of Butler (1986) that drawdown during a pumping test can be considered as consisting of two components: one dependent on the properties of near-well material and one independent of such material. Earlier work of Wenzel (1942) and Jacob (1963) discusses steady and unsteady radial flow during a pumping test. Steady radial flow can be reinterpreted here as the period during which the rate of drawdown is independent of the properties of near-well material, i.e. that period during which eqn. (23) is applicable. In this and subsequent discussions, the term "near-well" refers to locations near an observation well.

Further insight can be gained by applying the Cooper-Jacob approach over a single $\log_{10}$ cycle in distance resulting in the following expression for transmissivity:

$$T_1 = \frac{2.3Q}{2\pi\Delta s_d} \quad (24)$$

where Δs_d = change in drawdown over a single $\log_{10}$ cycle in distance. This expression is simply a form of the Thiem equation (Bear, 1979), thus showing the equivalence of the Thiem and Cooper-Jacob approaches. Though first proven by Wenzel (1942), the transient nature of the Thiem equation has received little attention in the literature. Equation (24) demonstrates that the realm of applicability of the Thiem equation can be extended to include the transient, steady-flow conditions discussed above.

In the case of eqn. (21), it can be shown that, analogous to the denominator, the numerator of the bracketed expression can be approximated as $1/NR$. This allows reduction to a form that can be analytically transformed to the following expression:

$$s_2 = \frac{Q}{4\pi T_2} \ln \left(\frac{4T_2 t}{CS_2 r^2} \right) \quad (25)$$

Equation (25) is equivalent to the large-time solution (A.13) of Barker and Herbert (1982). Note that at large dimensionless times total drawdown in the matrix and its rate of change are a function solely of matrix properties. Consideration of small-time behavior, however, reveals that properties of the embedded disk do play a limited, transient role as shown by eqn. (17) and displayed in Fig. 4. The decreasing dependence of matrix drawdown on the properties of the embedded disk with time is analogous to behavior observed

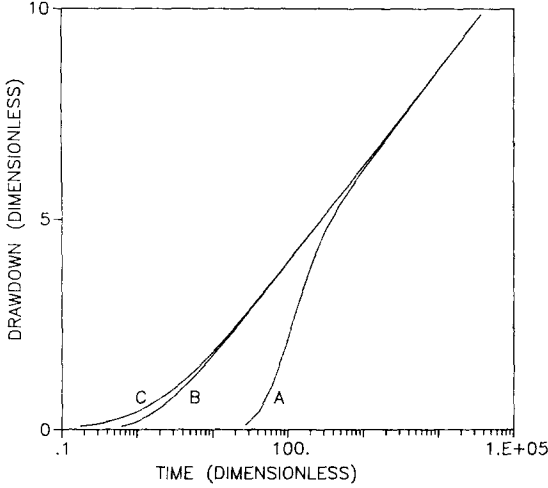


Fig. 4. Dimensionless drawdown ($4\pi s_2 T_2/Q$) versus time ($4T_2 t/S_2 r^2$) plot of numerically inverted solution ($R/r = 0.9$, $S_2/S_1 = 1.0$, $T_{1A}/T_2 = 0.01$, $T_{1B}/T_2 = 1$, $T_{1C}/T_2 = 100$).

in the vicinity of a large-diameter pumping well (Papadopoulos, 1967), and is a reflection of evolving flow conditions at the disk-matrix boundary. As with eqn. (22), application of the Cooper-Jacob approach through time will yield an expression equivalent to (23).

It should be emphasized that the large dimensionless time employed for nonuniform configurations, equivalent to AR , NR , Nr , $Ar < 0.1$ in Laplace space, is different from that used in the uniform case. In the nonuniform case, distances to both the observation point and the discontinuity must be considered in order to allow time for transients produced both by material in the vicinity of the observation point and by the discontinuity to be damped.

Implementation of solution for pumping-test analyses

Since one of the motivations for this work was to examine approaches for analysis of response data from a pumping test in a configuration that can be roughly conceptualized as pumping from a circular disk embedded in a matrix of differing properties, the discussion must now turn to the implementation of the above solution for parameter estimation. Some suggestions will be made here to improve the curve-matching methods of Bixel and Van Poollen (1967) and Şen (1987).

The objective of the analysis is to estimate T_1 , S_1 , T_2 , S_2 , and R from the response data. If the boundary R is at a relatively large distance, then, at early times, the convergent properties of modified Bessel functions of the second kind of order 0 and 1 can be employed to simplify eqn. (16) to:

$$\bar{s}_1 = \left(\frac{Q}{2\pi T_1} \right) \frac{K_0(Nr)}{p} \quad (26)$$

which is the Laplace-spaced form of the solution of Theis (1935). Standard techniques for uniform aquifers can therefore be utilized to estimate T_1 and S_1 from the early-time drawdown in the disk. If several observation points are in the disk, an approach based upon eqn. (24) can be employed to estimate T_1 at large dimensionless times. The time at which the effects of the discontinuity are first noted can be used to calculate R by employing a simple curve-matching approach analogous to that developed by Lohman (1972) for estimating distance to a recharge or discharge boundary. After the transient effects of the discontinuity have dissipated, eqn. (23) can be utilized to estimate T_2 . A Cooper-Jacob analysis through time can also be applied to calculate the storage coefficient, resulting in the following equation for S_2 :

$$S_2 = \frac{4 T_2 t_0}{CR^2 \left(\frac{r}{R} \right)^{2T_2/T_1}} \quad (27)$$

where $t_0 = x$ intercept of the large-time, straight-line portion of the semilog drawdown plot.

For an observation point located in the matrix, T_2 and S_2 can be simply calculated from eqn. (25) using the methods of Cooper and Jacob. The properties of the disk, however, cannot be readily calculated from drawdown in the matrix.

One of the difficulties encountered in computing T_2 and S_2 from drawdown in the disk is choosing when to apply eqns. (23) and (27). If care is not used, the chosen interval may still be in the transition period due to the discontinuity resulting in a spurious value for the parameters. In order to help define the end of the transition period, a plotting technique described by Bourdet et al.

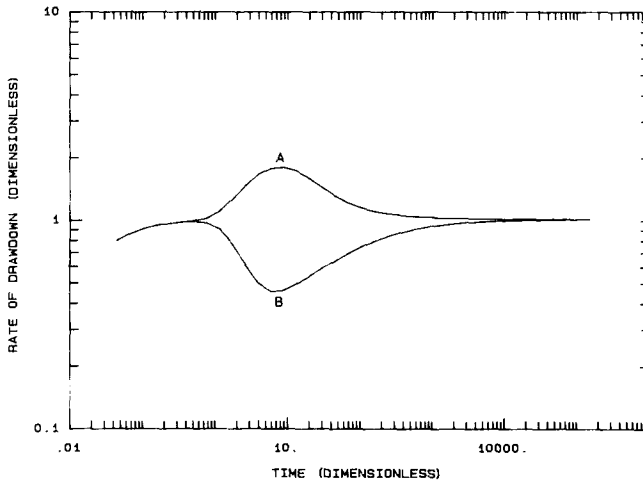


Fig. 5. Dimensionless plot of the rate of drawdown multiplied by time $[(\partial s_D / \partial t_D) t_D]$ versus time, illustrating behavior during the transition period (A and B as defined in text; s_D = dimensionless drawdown, t_D = dimensionless time).

(1983a, b) can be used. This approach employs a dimensionless plot of the rate of drawdown multiplied by time versus time, which results in a horizontal line when the transition period has passed, as shown in Fig. 5. Finite-difference approximations of the rate of drawdown can be employed to roughly estimate the bounds of the transition zone from field data, following Bourdet et al. (1983b).

It should be noted that these analytical techniques are quite useful for removing the dependence of aquifer parameters on drilling methodology, as the effect of the skin created by the different drilling approaches (the circular disk in this configuration) can be easily removed if the pumping test is carried out for a large enough time. The dependence of aquifer parameters on the diameter of the pumping well, which has often been noted in field studies (McKown et al., 1987), can also be removed in a similar fashion.

Implications for pumping tests in nonuniform aquifers

As stated earlier in this article, of greater importance than the use of the solution to (1)–(7) to derive methods of parameter estimation is the use of this solution to develop a better general understanding of flow behavior in nonuniform units during pumping tests. There are three major issues worthy of discussion in this regard: (1) the changing dependence of drawdown on near-well properties; (2) observation-well placement; and (3) drawdown behavior during the transition period caused by the discontinuity.

From the results of a numerical study of pumping tests in nonuniform aquifers, Butler (1986) hypothesized that drawdown during a pumping test consists of a component dependent on the properties of the material near the observation well and a component dependent on the properties of the material through which the front of the cone of depression is currently passing. It was conjectured that with time the additional contribution to total drawdown of the component dependent on near-well properties would go to zero. Equation (23) supports this hypothesis, as it is shown that the change in drawdown at large times is independent of near-well properties. A technique that employs the change in drawdown at large time, such as the Cooper–Jacob procedure, therefore allows estimation of the properties of the material through which the front of the cone of depression passed during the analyzed interval. A technique such as the Theis log–log curve-matching approach that involves total drawdown (i.e. both components), on the other hand, provides an estimate that is a weighted average of near-well and far-field properties. This differing emphasis on the two components of drawdown is the basis of the fundamental difference between pumping-test parameters, calculated using different analytical approaches, that was demonstrated by Butler (1986). It is of interest to note that the Cooper–Jacob method is as valid in nonuniform units of the type considered here as it is in uniform units, although the dimensionless time criteria are defined differently. The same cannot be said for the traditional curve-matching approaches, which provide a weighted average of unclear

significance. Further discussion of the fundamental differences between the curve-matching and Cooper–Jacob approaches occurs in Butler (1986).

One final point of interest concerning the Cooper–Jacob approach is the physical interpretation of the infinite-series truncation that is performed in both the uniform and nonuniform cases in order to obtain the linear relationships that yield eqn. (23). In the uniform case, little physical significance can be readily attached to the truncation. Jacob (1963) employed the relative rates of temporal and spatial drawdown to justify the approximation. In the nonuniform case, however, it is clear that the truncation is responsible for the removal of additional contributions of near-well materials to total drawdown, as the properties of the disk would otherwise play a role in the determination of the Δs of eqn. (23). The physical interpretation of the Cooper–Jacob truncation as having removed the additional contributions of near-well materials is of little significance in the uniform case. This explanation, however, is of great importance in nonuniform units since, in an analysis based on eqn. (23), the assumption of uniformity is only required for the material through which the front of the cone of depression is currently passing, a much less restrictive condition than in the curve-matching techniques.

The second major issue regarding pumping tests in nonuniform aquifers concerns observation-well placement. Consideration of eqns. (23)–(27) yields some insight into observation-well placement in nonuniform aquifers. An observation well placed near the pumping well should allow a fairly complete record of the properties of the material through which the front of the cone of depression has passed during a pumping test to be established. An observation well at a distance from the pumping well, on the other hand, will provide little information concerning properties of material between the two wells. In both cases, the aquifer volume over which parameters are estimated increases with the time of the analysis. Estimates of a more local scale can be obtained by using drawdowns from adjacent observation wells in an analysis based on eqn. (24). Although these statements are only strictly valid for an analysis of flow in a radially symmetric configuration, the general underlying principle, i.e. the concept of drawdown discussed in the previous paragraphs, is applicable to a much wider class of nonuniform units.

The third major issue concerning pumping tests in nonuniform units involves drawdown behavior during the transition period caused by the discontinuity. Fenske (1984) discussed this issue with respect to the transition due to a linear discontinuity. The nature of behavior in the transition period is clearly a function of the relative diffusivity of the two materials. A large matrix diffusivity relative to that of the disk will result in a transition characterized by a peaked rate of drawdown plot as illustrated by curve A of Fig. 5. The high diffusivity of the matrix results in increased drawdown in the disk until the gradient across the discontinuity has built up to the point at which sufficient flow across the boundary can occur. Similarly, a lower diffusivity in the matrix will result in a transition characterized by the rate of drawdown plot exhibiting a trough-like appearance as illustrated by curve B of Fig. 5. When the diffu-

sivity remains constant across the boundary, the rate of drawdown in the transition period will change monotonically and will not display the peaked behavior observed in the case of differing diffusivities. It should be emphasized that this general description of behavior in the transition period holds regardless of the specific nature of the discontinuity.

CONCLUSIONS

The derivation of a solution to the problem of flow in response to pumping from a well at the center of a circular disk, embedded in a matrix of differing properties, allows techniques to be proposed for calculating material properties in aquifers whose structure can be roughly represented by this simple configuration. More importantly, the large-time form of this solution provides significant insight into flow behavior during a pumping test for a wide class of nonuniform aquifers. The solution demonstrates that drawdown during a pumping test can be considered to consist of two components that are dependent and independent of near-well properties, respectively. This interpretation of pumping-test drawdown allows some conclusions to be drawn concerning the difference between parameters calculated using curve-matching and semilog straight-line analytical approaches. In addition, some conclusions are reached concerning the physical significance of the infinite-series truncation suggested by Cooper and Jacob (1946) and the transient nature of the Thiem equation. Finally, some general recommendations for observation-well placement in nonuniform aquifers are also proposed. The results of this work demonstrate that pumping tests in nonuniform aquifers are a worthwhile approach, even without a large number of observation wells, as much information about large-scale property variations in the subsurface can be obtained. Present work is examining the possibility of applying analytical approaches to more complex configurations in hopes of bettering understanding of pumping-test behavior under more general nonuniform conditions.

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